

August 6, 2026

# Chromebooks Are the New Thinkpads

## Turning E-Waste into a Daily Driver

Owen Zimmerman, Fyra Labs

# Hey, I'm Owen

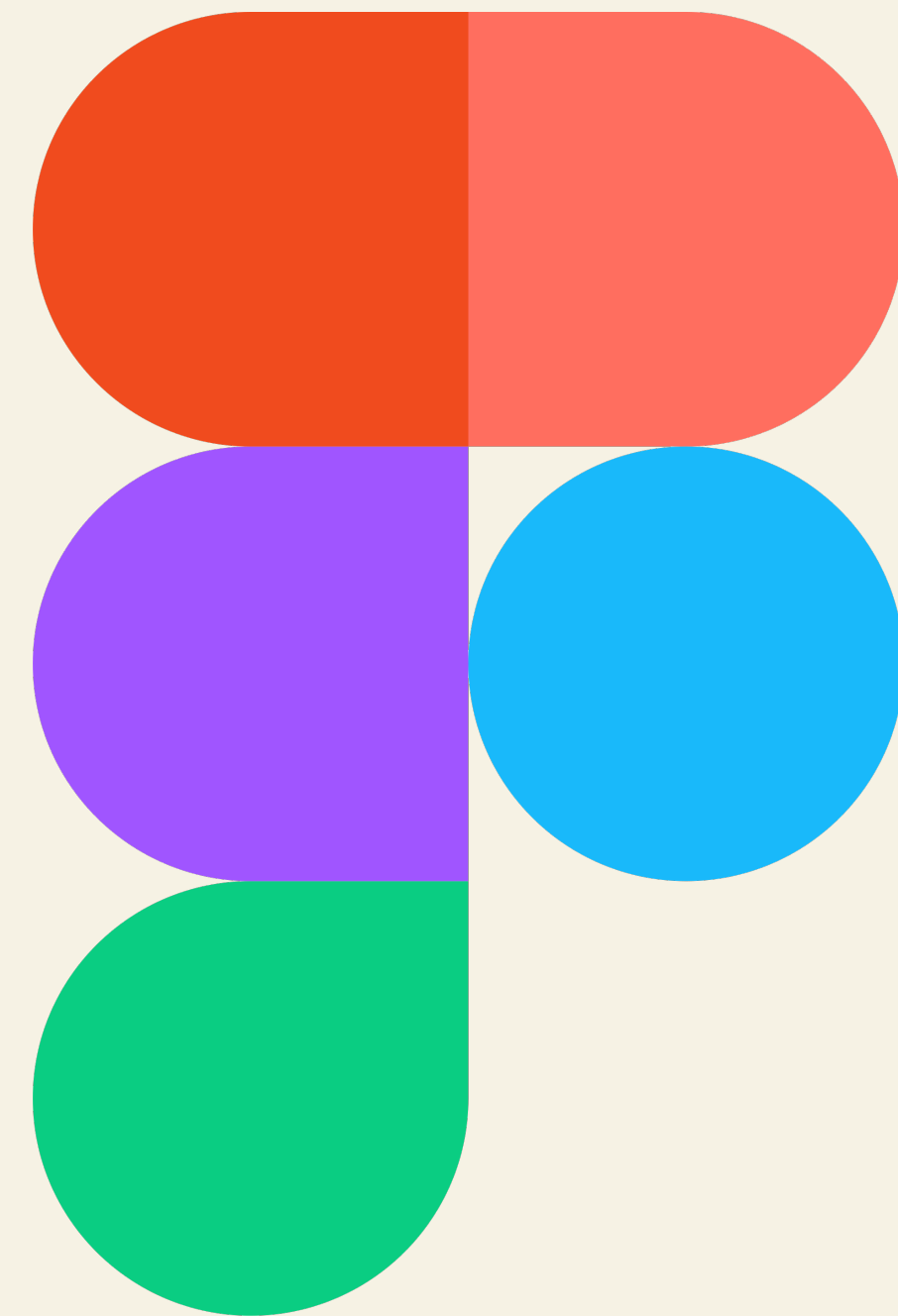
If you've already seen my bio, here's a recap:

- Chief Business Officer @ *Fyra Labs*
- Co-Founder @ *Fyra Stack*
- A Lead Maintainer of the *Terra repository*
- Also a Lead Maintainer of *Ultramarine*
- *Open Gaming Collective* org member
- Huge Star Wars and Movies fan
- Owns 100 Computers, 50 Chromebooks



# Proprietary Software

Per conference rules I am  
required to point out any  
proprietary software in use.



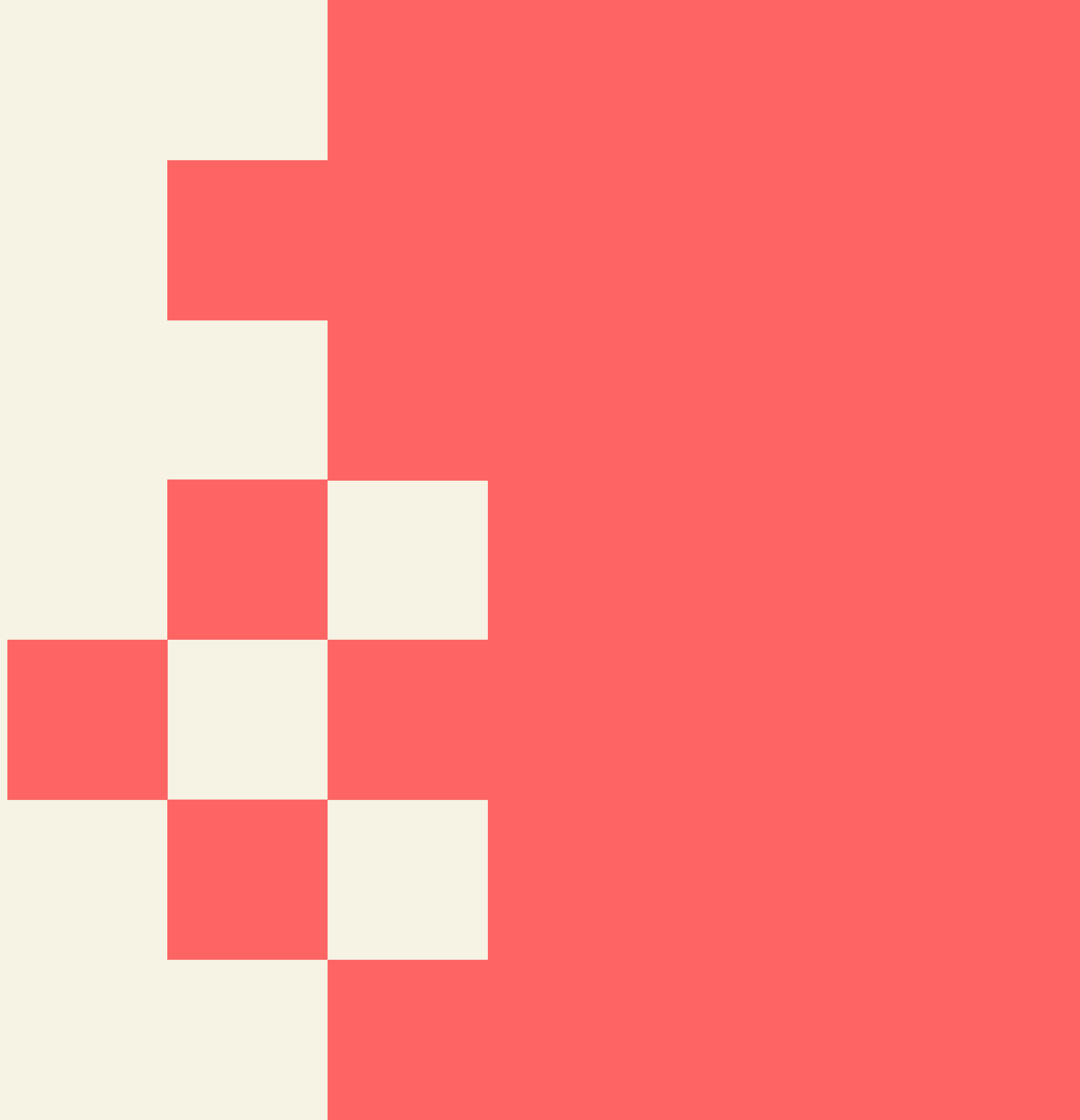
# Agenda

**How it Works**

**Reducing E-Waste**

**Demo**

**Questions**



# How it Works

# Alternative Firmware

## 01 mrchromebox Coreboot

mrchromebox maintains a coreboot fork for x86\_64 Chrome devices, enabling full UEFI firmware support.

## 02 Firmware Write Protect

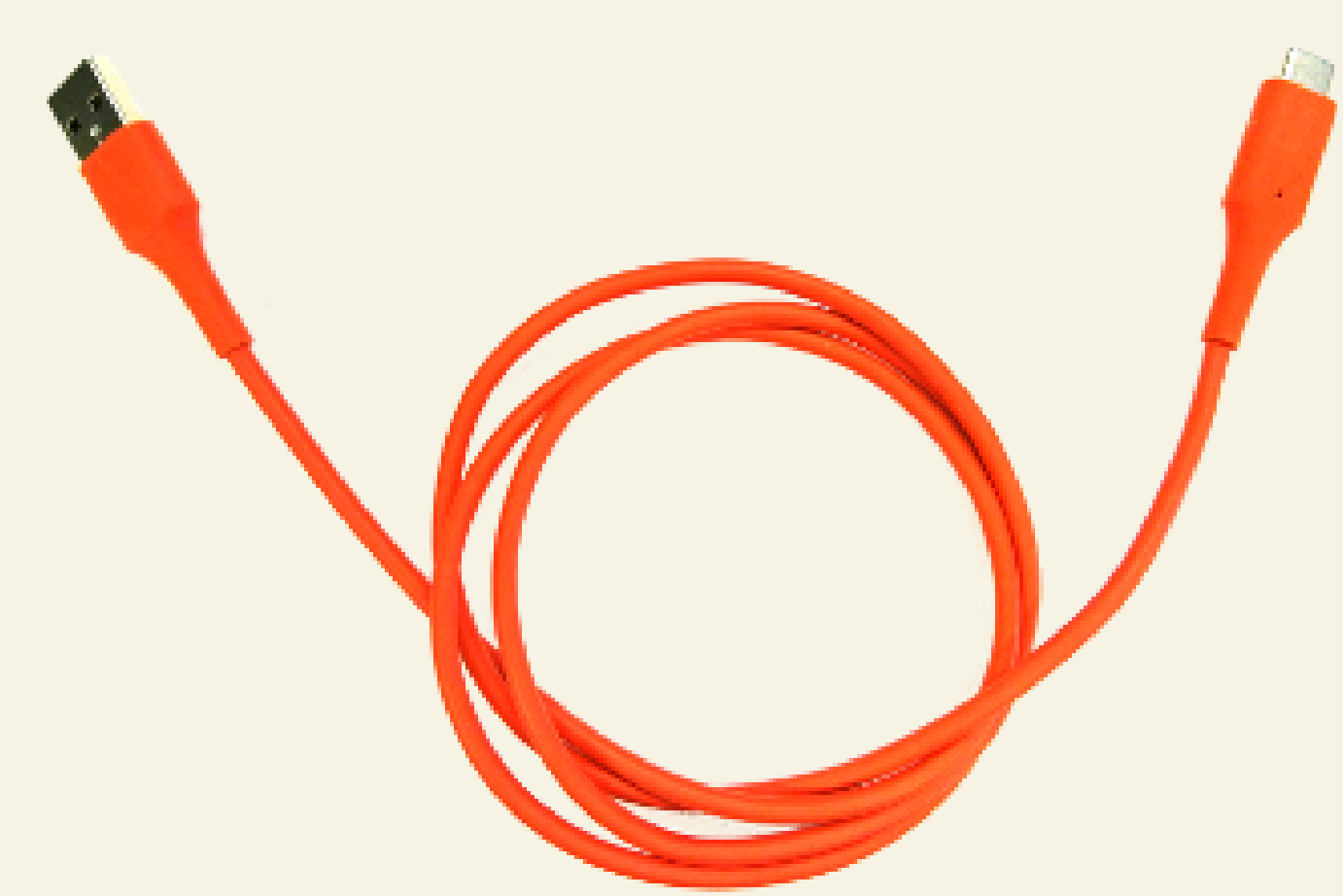
A physical firmware writing protection method present on all Chrome devices. Must be disabled in order to flash firmware.

## 03 Disabling Firmware WP

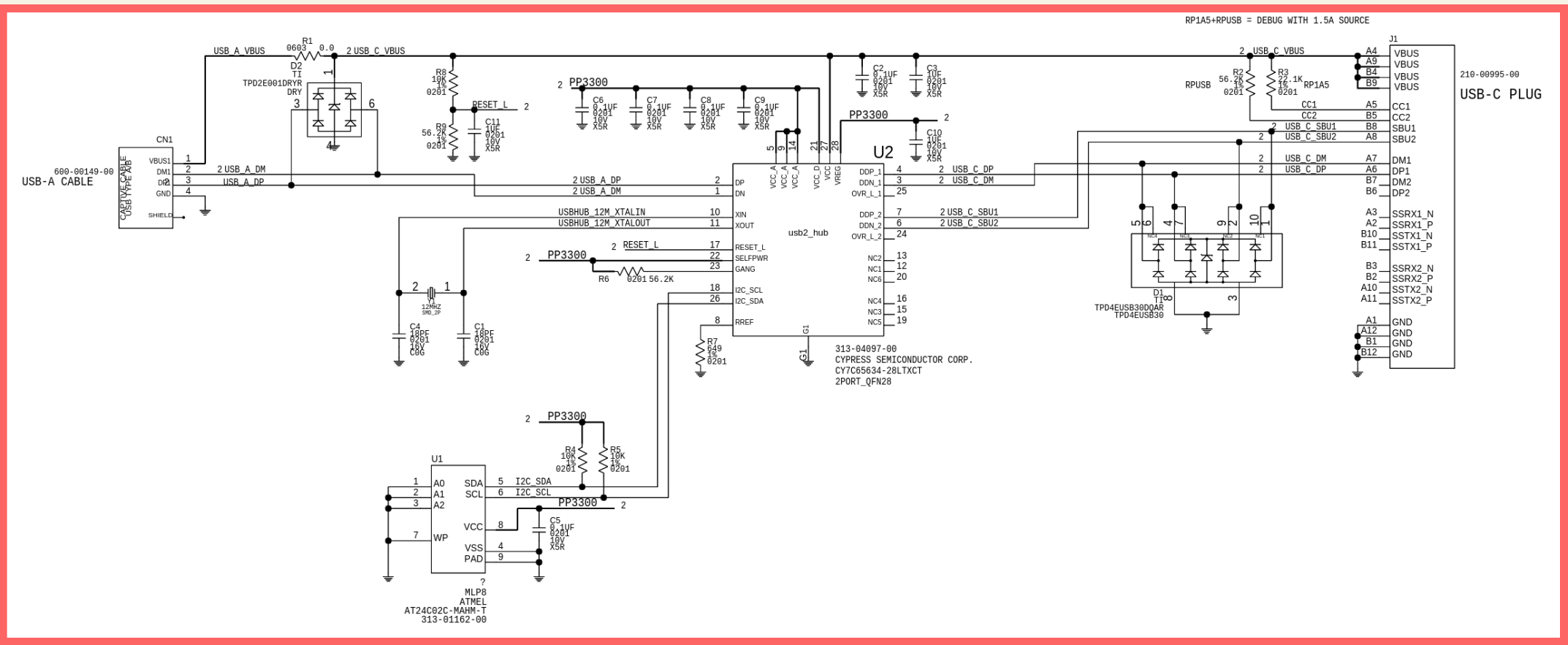
Older x86\_64 Chrome devices can have their WP disabled via hardware switches, though newer ones need a SuzyQable to flash.

# Closed-Case Debugging

Modern Chromebooks can be interfaced with Closed-Case Debugging (CCD). This enables tool-less EC communication, firmware flashing, unbricking, and more.



A SuzyQable  
Image Courtesy: Chromium Developers



SuzyQable Schematic  
Image Courtesy: Chromium Developers



A SuzyQable Built By WeirdTreeThing  
Image Courtesy: Me

# Fyra Labs SuzyQ Board

We partnered with the Chruultrabook Project to build a SuzyQ PCB.  
We are now the only company who sells one, as Google discontinued their SuzyQables.



Our SuzyQ Board  
Image From Fyra Shop



Packaging The First Production Run  
Courtesy: Owen Zimmerman

# Linux

## 01 Chrultrabook Project

The Chrultrabook project ensures Linux (and other alternative OSs) work on Chromebooks, they do everything from driver and audio development, firmware development, utility and application development, community support, and more.

## 02 Ultramarine Linux

Ultramarine Linux has a large focus on hardware enablement, with Chromebooks being a big part of that.

We package many Chromebook modding utilities, help with needed upstreaming (Fedora kernel configs, Alsa configs), co-maintain a stock firmware bootloader, and are working on x86\_64 and aarch64 images...

## 03 Other distros

As of 2024, the officially recommended distros by the Chrultrabook Project are:

- Arch Linux or EndeavourOS
- Fedora or Ultramarine Linux
- openSUSE Tumbleweed
- Pop!\_OS

# Reducing E-Waste

# Reducing E-waste in Our Community

- Work with Free Geek Twin Cities to provide technical expertise for reselling end-of-life Chromebooks
- Helped a Best Buy Teen Tech Center use forgotten Chromebooks



Getting Chromebooks Ready For Resale  
Image Courtesy: Jonah

# Reducing E-Waste in YOUR Community

- Volunteer at local E-Waste and recycling center facilities
- Explore schools/educational institutions, help out with computer programs
- Gather local unwanted Chromebooks and other devices

# Live Demo

**Demo gods smile upon me**

# Questions?

# Land Acknowledgment

Go to [fyra.to/vancouver](https://fyra.to/vancouver) for our land acknowledgment.



A Prayer Flag At Bdóte

Image Courtesy Wikimedia Commons & Lorie Shaul

Thank You

# Thank You!

**Get a SuzyQ → [fyra.to/suzyq](https://fyra.to/suzyq)**

**Fyra Labs @ FOSSY → [fyra.to/fossy26](https://fyra.to/fossy26)**

**More from Owen → [github.com/owen-sz](https://github.com/owen-sz)**

## A Fyra Labs Joint

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